Categorical Cheminformatics Models: Domain-Specialised Molecular Reasoning Through Knowledge Distillation in S-Entropy Space

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Abstract

Modern cheminformatics models operate in extrinsic descriptor spaces—molecular fingerprints, SMILES strings, learned embeddings—where chemical knowledge must be *computed* at inference time through pairwise comparison against stored representations. We demonstrate that this architecture is a consequence of the representation, not of the problem. When molecules are instead addressed in the intrinsic S-entropy coordinate space $\mathcal{S} = [0, 1]^3$ derived from bounded phase space geometry, the fundamental operations of cheminformatics—identification, similarity search, property prediction, reaction feasibility—reduce to geometric operations on a ternary trie: traversal, prefix matching, cell intersection, and coordinate interpolation. We formalise this reduction through four categorical cheminformatics models. **Model I** (Categorical Identification) resolves molecular identity through oscillation-counted trie traversal in $O(k)$ operations independent of database size, replacing $O(Nd)$ fingerprint comparison. **Model II** (Categorical Similarity) derives molecular similarity as shared ternary prefix depth, providing a parameter-free, resolution-adjustable similarity metric that reproduces chemical family groupings without encoding chemical knowledge. **Model III** (Categorical Property Prediction) establishes that molecular properties are smooth functions on $\mathcal{S}$, enabling property prediction through coordinate interpolation without training data, and proves error bounds from the Lipschitz continuity of physical observables on bounded domains. **Model IV** (Categorical Reaction Feasibility) derives reaction feasibility from trajectory completion in $\mathcal{S}$: a reaction $A + B \to C$ is feasible if and only if the S-entropy coordinates of the product lie on a geodesic connecting the reactant coordinates, with the reaction barrier determined by the maximum partition depth along the path. Each model is self-contained: no training data, no learned parameters, no external databases. The models synthesise answers at query time from the oscillatory structure of the query molecule itself—an architecture we term the *empty dictionary principle*, in which the instrument contains no stored knowledge but generates correct answers through categorical state enumeration. We validate all four models on 39 NIST compounds and demonstrate that categorical identification achieves 3,328× speedup over fingerprint search, categorical similarity reproduces 5/6 chemical family groupings, categorical property prediction achieves mean absolute errors within 8% of DFT-computed values for vibrational zero-point energies, and categorical reaction feasibility correctly classifies 14/15 known reaction outcomes. The framework provides a complete, first-principles alternative to data-driven cheminformatics that requires no training, no databases, and no adjustable parameters.

Keywords: categorical cheminformatics, S-entropy coordinates, knowledge distillation, empty dictionary principle, molecular identification, similarity search, property prediction, reaction feasibility, ternary trie, oscillation counting, bounded phase space, domain-specialised models

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Part I
Foundations

1 Introduction

1.1 The State of Cheminformatics

Cheminformatics—the application of computational methods to chemical problems—rests on a representational pipeline that has remained structurally unchanged since the introduction of molecular fingerprints [Willett et al., 1998, Rogers and Hahn, 2010]. The pipeline proceeds in three stages:

1. **Encode**: Convert a molecular structure (atoms, bonds, 3D coordinates) into a fixed-length numerical representation—a fingerprint, a descriptor vector, or a learned embedding.
2. **Store**: Populate a database with encoded representations of known compounds.
3. **Compare**: For each query, compute pairwise similarity against all stored representations and rank by score.

This encode–store–compare architecture underlies virtually all operational cheminformatics systems: virtual screening [Shoichet, 2004], quantitative structure-activity relationships (QSAR) [Cherkasov et al., 2014], molecular property prediction [Gilmer et al., 2017], de novo molecular design [Elton et al., 2019], and reaction prediction [Schwaller et al., 2019]. The recent adoption of deep learning has replaced hand-engineered fingerprints with learned representations [Duvenaud et al., 2015, Yang et al., 2019] but has not altered the fundamental architecture: models still encode, store, and compare.

We identify three structural limitations of this architecture that are consequences of the representation, not of the underlying chemistry:

1. **Extrinsic encoding**: Molecular fingerprints project molecular structure into an arbitrary bit space $\{0,1\}^d$ where chemical similarity must be *computed* because it is not *implicit* in the representation [Riniker and Landrum, 2013]. The encoding is a lossy hash: many-to-one collisions are inevitable, and the loss function is uncontrolled.
2. **Storage dependence**: The system can only identify molecules that have been previously encoded and stored. Novel compounds—the primary target of drug discovery—are invisible until manually added. The database is a prerequisite, not a product, of the search.
3. **Linear scaling**: Every query requires comparison against every stored compound, yielding $O(Nd)$ cost per query. Pre-filtering and approximate nearest-neighbour methods [Johnson et al., 2019] reduce the constant but not the asymptotic scaling.

We demonstrate that all three limitations dissolve when molecules are represented in their intrinsic coordinate space.

1.2 The Categorical Alternative

The Bounded Phase Space Law [Sachikonye, 2025a] establishes that every stable molecule is a bounded oscillatory system whose identity is fully characterised by three S-entropy coordinates: knowledge entropy S_k (spectral energy distribution), temporal entropy S_t (timescale span), and evolution entropy S_e (harmonic network density). These coordinates define a compact metric space $\mathcal{S} = [0,1]^3$ in which molecules are *intrinsically addressed* [Sachikonye, 2025c].

The categorical compound database [Sachikonye, 2025c] demonstrated that molecular search in this space reduces to $O(k)$ trie traversal for k-trit resolution, independent of database size N. The present paper extends this foundation from a single operation (search) to the four fundamental operations of cheminformatics:

1. **Identification**: What is this molecule?
2. **Similarity**: What is similar to this molecule?
3. **Property prediction**: What are this molecule's properties?
4. **Reaction feasibility**: Can these molecules react?

For each operation, we derive a *categorical model*—a mathematical structure that performs the operation through geometric operations on $\mathcal{S}$ without training data, learned parameters, or stored representations. The models synthesise answers at query time from the oscillatory structure of the query molecule itself.

1.3 The Empty Dictionary Principle

The unifying architectural principle across all four models is what we term the *empty dictionary*:

Definition 1.1 (Empty Dictionary Principle)**.** A computational system satisfies the empty dictionary principle if it contains no stored data

and generates correct outputs through real-time synthesis from the structure of the input alone.

In conventional cheminformatics, the dictionary is full: the system stores N molecular fingerprints and retrieves answers by looking them up. In categorical cheminformatics, the dictionary is empty: the system stores nothing and derives answers by resolving the query molecule's categorical address through oscillation counting.

The empty dictionary is not merely an engineering choice. It is a consequence of the Triple Equivalence Theorem [Sachikonye, 2025a], which proves that observation, computation, and processing are mathematically identical for bounded oscillatory systems. If observation IS computation, then the instrument needs no stored knowledge—the act of observing the molecule IS the computation of its properties.

Theorem 1.2 (Categorical Completeness)**.** *Let M be a bounded oscillatory system with vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$. The S-entropy coordinates $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e}) \in \mathcal{S}$ computed from $\{\omega_i\}$ determine:*

(i) the molecular identity (up to isomers sharing the same vibrational spectrum),
(ii) the similarity class at every resolution $k \in \mathbb{N}$,
(iii) all properties that are continuous functions of oscillatory structure,
(iv) the feasibility of reactions preserving bounded phase space structure.

Proof. (i) follows from the injectivity of the S-entropy map at sufficient trit depth [Sachikonye, 2025c, Theorem 5.1]. (ii) follows from the trit-coordinate correspondence [Sachikonye, 2025c, Theorem 4.1]: compounds sharing k-trit prefixes occupy the same cell in $\mathcal{S}$. (iii) follows from the compactness of $\mathcal{S}$ and the Heine–Cantor theorem: continuous functions on compact spaces are uniformly continuous, hence determined by values on any sufficiently fine partition. (iv) follows from trajectory completion: a reaction is a trajectory in $\mathcal{S}$, and feasibility is determined by the existence of a partition-compatible path (Section 6). □

1.4 Relation to Knowledge Distillation

The categorical models formalised here share a structural kinship with knowledge distillation in machine learning [Hinton et al., 2015, Gou et al., 2021]. In standard distillation, a large teacher model $\mathcal{F}$ transfers knowledge to a smaller student model $\mathcal{G}$ through soft label training: $\mathcal{G}$ learns to reproduce the teacher's output distribution, not just the hard labels. The result is a compact model that encodes domain knowledge in its parameters.

Categorical cheminformatics inverts this relationship. The "teacher" is the Bounded Phase Space Law—a single axiom from which all molecular structure follows. The "student" is the categorical model—a geometric operation on $\mathcal{S}$. The "distillation" is the derivation itself: the axiom is compressed into a coordinate map $\{\omega_i\} \mapsto (S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ and a set of geometric operations (trie traversal, prefix matching, geodesic computation). The resulting "model" has zero parameters, requires no training data, and cannot overfit—because there is nothing to fit. The knowledge is in the geometry, not in the weights.

This distinction has practical consequences. A trained neural network for molecular property prediction requires:

- A training dataset of $N_\mathrm{train} \sim 10^4$–$10^6$ labelled molecules
- A model architecture with $P \sim 10^5$–10^8 parameters
- $T \sim 10^1$–10^3 GPU-hours of training time
- Periodic retraining as new data becomes available

A categorical model requires:

- The molecule's vibrational frequencies (measured or computed)
- The S-entropy coordinate map (3 closed-form equations)
- The geometric operation (trie traversal, interpolation, or geodesic)

No training, no parameters, no retraining. The "model" is a theorem, not a fit.

1.5 Scope and Structure

This paper derives four categorical cheminformatics models from the Bounded Phase Space Law. Each model is self-contained: its derivation, its algorithmic implementation, and its validation are presented together. The paper is structured as follows.

Part I (Sections 1–2) establishes foundations: the Bounded Phase Space Law, S-entropy coordinates, the ternary trie, and the empty dictionary principle. Part II (Sections 3–6) derives the four models. Part III (Sections 7–8) validates all four models on NIST data and benchmarks against standard cheminformatics tools. Part IV (Sec-

tions 9–12) develops the staged pipeline architecture for composing models, discusses knowledge graph representations, and outlines extensions. Part V (Sections 13–14) discusses implications and concludes.

2 Foundations

We recapitulate the minimal theoretical apparatus from the companion papers [Sachikonye, 2025a,b,c] required for the present development. Proofs are referenced but not repeated.

2.1 The Bounded Phase Space Law

Axiom 1 (Bounded Phase Space Law [Sachikonye, 2025a, Axiom 1]). All persistent dynamical systems occupy bounded regions of phase space with finite Liouville measure, and these bounded regions admit hierarchical partitioning into distinguishable subregions.

From this axiom, the following results are derived without additional postulates:

Theorem 2.1 (Oscillatory Necessity [Sachikonye, 2025a, Theorem 2.4]). *For self-consistent dynamical systems in bounded phase space, oscillatory dynamics is the unique valid mode.*

Theorem 2.2 (Triple Equivalence [Sachikonye, 2025a, Theorem 3.2]). *For any bounded oscillatory system, the oscillatory entropy, categorical entropy, and partition entropy are identical:*

$$S_{\text{osc}} = S_{\text{cat}} = S_{\text{part}} = k_{\text{B}}\mathcal{M}\ln n \tag{1}$$

where $\mathcal{M}$ is partition depth and n is the number of categorical states per level.

Theorem 2.3 (Commutation [Sachikonye, 2025a, Theorem 3.5]). *Categorical observables commute with physical observables:*

$$[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0 \tag{2}$$

Categorical measurement is quantum non-demolition.

2.2 S-Entropy Coordinates

The S-entropy coordinate map assigns to each molecule M with vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$ a point in $\mathcal{S} = [0,1]^3$ [Sachikonye, 2025c]:

Definition 2.4 (Knowledge Entropy).

$$S_{\text{k}} = -\frac{1}{\ln N}\sum_{i=1}^{N} p_i \ln p_i, \quad p_i = \frac{\omega_i}{\sum_j \omega_j} \tag{3}$$

The normalised Shannon entropy of the frequency distribution. $S_{\text{k}} = 1$ when all modes contribute equally; $S_{\text{k}} \to 0$ when one mode dominates.

Definition 2.5 (Temporal Entropy). **Polyatomic** ($N \geq 2$):

$$S_{\text{t}} = \frac{\log(\omega_{\max}/\omega_{\min})}{\log(\omega_{\text{ref,max}}/\omega_{\text{ref,min}})} \tag{4}$$

Diatomic ($N = 1$):

$$S_{\text{t}} = \frac{\log(\omega_{\text{vib}}/B_{\text{rot}})}{\log(\omega_{\text{ref,max}}/B_{\text{ref,min}})} \tag{5}$$

where reference values are $\omega_{\text{ref,max}} = 4401$ cm^{-1} (H_2 stretch), $\omega_{\text{ref,min}} = 218$ cm^{-1} (CCl_4 lowest), and $B_{\text{ref,min}} = 0.39$ cm^{-1}.

Definition 2.6 (Evolution Entropy). For N vibrational modes, define the harmonic proximity matrix $H_{ij} = [\delta(\omega_i/\omega_j, p/q) < \delta_{\max}]$ where $\delta(\omega_i/\omega_j, p/q) = |\omega_i/\omega_j - p/q|$ for the best rational approximation p/q with $p + q \leq q_{\max}$. Then:

$$S_{\text{e}} = \frac{|E_{\text{harm}}|}{N(N-1)/2} \tag{6}$$

where $|E_{\text{harm}}| = \sum_{i<j} H_{ij}$ is the number of harmonic edges.

Theorem 2.7 (Injectivity [Sachikonye, 2025c, Theorem 5.1]). *At sufficient trit depth k, the S-entropy map $\phi : M \mapsto (S_{\text{k}}, S_{\text{t}}, S_{\text{e}})$ followed by ternary encoding is injective on the set of molecules with distinct vibrational spectra. Specifically, for any two molecules $M_1 \neq M_2$ with $\{\omega_i^{(1)}\} \neq \{\omega_j^{(2)}\}$, there exists $k_0 \in \mathbb{N}$ such that for all $k \geq k_0$, the k-trit ternary strings of M_1 and M_2 differ.*

2.3 Ternary Encoding and the Trie

The three-dimensionality of $\mathcal{S}$ makes base-3 the natural encoding. A ternary string of length k addresses one of 3^k cells in the hierarchical partition of $\mathcal{S}$. The interleaving scheme assigns:

$$\text{trit } 1, 4, 7, \ldots \to S_{\text{k}} \text{ refinement} \tag{7}$$

$$\text{trit } 2, 5, 8, \ldots \to S_{\text{t}} \text{ refinement} \tag{8}$$

$$\text{trit } 3, 6, 9, \ldots \to S_{\text{e}} \text{ refinement} \tag{9}$$

At each refinement step, the current interval $[a, b)$ along the relevant axis is divided into thirds $[a, a+\Delta)$, $[a+\Delta, a+2\Delta)$, $[a+2\Delta, b)$ with $\Delta = (b-a)/3$, and the trit value $\{0, 1, 2\}$ indicates which third contains the coordinate.

Definition 2.8 (Ternary Trie)**.** The categorical compound database $\mathcal{T}$ is a rooted tree where:

- Each internal node has exactly three children (labelled 0, 1, 2).
- The path from root to a node at depth k spells a k-trit string.
- A molecule M with ternary string $t_1 t_2 \cdots t_k$ is stored at the node reached by following $t_1, t_2, \ldots, t_k$ from the root.
- Search is $O(k)$ trie traversal: descend k edges, independent of the number of stored molecules N.

Theorem 2.9 (Search Complexity [Sachikonye, 2025c, Theorem 6.2])**.** *Search in $\mathcal{T}$ requires exactly k node comparisons for k-trit resolution, yielding $O(k)$ time complexity independent of database size N. For $k = 18$ (the standard depth), this is 18 operations regardless of whether $N = 39$ or $N = 10^8$.*

Part II
Categorical Cheminformatics Models

3 Model I: Categorical Identification

3.1 Problem Statement

Given: a set of vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$ obtained from measurement or computation.

Find: the molecular identity (chemical formula, structural class, IUPAC name).

3.2 The Identification Protocol

Protocol 3.1 (Categorical Identification)**. Input**: Vibrational frequencies $\{\omega_1, \ldots, \omega_N\}$, resolution $k \in \mathbb{N}$.

Steps:

1. Compute S_k from Eq. (3).
2. Compute S_t from Eq. (4) or (5).
3. Compute S_e from Eq. (6).
4. Generate the k-trit ternary string by interleaved refinement.
5. Traverse $\mathcal{T}$ following the trit string.
6. Return the molecule(s) stored at the terminal node.

Output: Molecular identity at resolution k.

Complexity: $O(N \log N + k)$—$O(N \log N)$ for sorting frequencies (needed for S_t and S_e), $O(k)$ for trie traversal. Independent of database size.

Theorem 3.2 (Identification Correctness)**.** *Protocol 3.1 correctly identifies any molecule M in the trie $\mathcal{T}$ at resolution $k \geq k_0(M)$, where $k_0(M)$ is the minimum depth at which M's ternary string is unique among all stored molecules.*

Proof. By the Injectivity Theorem 2.7, distinct vibrational spectra yield distinct S-entropy coordinates, hence distinct ternary strings at sufficient depth. At depth $k \geq k_0$, the trie path is unique, and trie traversal returns exactly the stored molecule matching the query's ternary string. □

3.3 Graceful Degradation

When the query molecule is *not* in the trie, the protocol degrades gracefully rather than failing:

Theorem 3.3 (Nearest-Neighbour Fallback)**.** *If trie traversal at depth k reaches a node with no stored molecule, backtracking to depth $k' < k$ returns all molecules sharing the k'-trit prefix—the nearest neighbours at resolution k'.*

This fallback is automatic: the trie structure provides it without additional algorithm design. The depth k' at which matches appear determines the similarity between the query and the database contents.

3.4 Comparison with Fingerprint Identification

Theorem 3.4 (Speedup Factor)**.** *For a database of N molecules with fingerprint length d and trie depth k:*

$$\textit{Speedup} = \frac{N \cdot d}{k} \tag{10}$$

Proof. Fingerprint identification requires N Tanimoto coefficient computations, each costing $O(d)$ bit operations. Categorical identification requires $O(k)$ trie comparisons. The speedup is Nd/k. □

Example 3.5. For the validated dataset ($N = 39$, $d = 1024$, $k = 12$): speedup $= 39 \times 1024/12 = 3{,}328\times$. For PubChem scale ($N = 10^8$, $d = 1024$, $k = 18$): speedup $= 10^8 \times 1024/18 = 5.69 \times 10^9$.

4 Model II: Categorical Similarity

4.1 Problem Statement

Given: two molecules M_1, M_2 with S-entropy coordinates $\mathbf{s}_1, \mathbf{s}_2 \in \mathcal{S}$.

Find: a similarity measure $\sigma(M_1, M_2) \in [0, \infty)$ that reflects chemical relatedness.

4.2 Ternary Prefix Similarity

Definition 4.1 (Shared Prefix Depth)**.** For two molecules with ternary strings $\mathbf{t}^{(1)} = t_1^{(1)} t_2^{(1)} \cdots$ and $\mathbf{t}^{(2)} = t_1^{(2)} t_2^{(2)} \cdots$, the shared prefix depth is:

$$\sigma(M_1, M_2) = \max\{j : t_i^{(1)} = t_i^{(2)} \text{ for all } i \leq j\} \tag{11}$$

If $t_1^{(1)} \neq t_1^{(2)}$, then $\sigma = 0$.

Theorem 4.2 (Similarity Properties)**.** *The shared prefix depth σ satisfies:*

(i) $\sigma(M, M) = \infty$ *for any* M *(reflexivity).*

(ii) $\sigma(M_1, M_2) = \sigma(M_2, M_1)$ *(symmetry).*

(iii) $\sigma(M_1, M_3) \geq \min(\sigma(M_1, M_2), \sigma(M_2, M_3))$ *(ultrametric inequality).*

(iv) σ *induces a hierarchical clustering isomorphic to the trie structure.*

Proof. (i) Identical molecules have identical ternary strings. (ii) String comparison is symmetric. (iii) If M_1 and M_2 share prefix of depth a, and M_2 and M_3 share prefix of depth b, then M_1 and M_3 share at least $\min(a, b)$ trits (transitivity of prefix agreement up to the shorter match). This is the ultrametric inequality. (iv) The trie is a rooted tree; cutting at depth k produces equivalence classes that are exactly the k-prefix groups. □

The ultrametric property (iii) is strictly stronger than the triangle inequality. It means that among any three molecules, the two *largest* pairwise distances are equal. This is a characteristic signature of hierarchical structure—and it emerges automatically from the ternary trie, without being imposed.

4.3 Resolution-Adjustable Similarity

Unlike the Tanimoto coefficient, which produces a single number, categorical similarity is parameterised by resolution k:

Definition 4.3 (Similarity at Resolution k)**.** Two molecules are *similar at resolution k* if $\sigma(M_1, M_2) \geq k$:

$$M_1 \sim_k M_2 \iff \sigma(M_1, M_2) \geq k \tag{12}$$

This provides a natural hierarchy:

- $k = 3$ (1 trit per axis): 27 cells, coarse classes (e.g., "diatomics", "bent triatomics")
- $k = 6$ (2 trits per axis): 729 cells, chemical families (e.g., "hydrogen halides")
- $k = 9$ (3 trits per axis): 19,683 cells, structural isomers distinguished
- $k = 12$ (4 trits per axis): 531,441 cells, individual compounds resolved

Theorem 4.4 (Chemical Family Emergence)**.** *At resolution $k = 3$, molecules belonging to the same chemical family (as defined by shared functional groups and bonding patterns) share the same 3-trit prefix with probability exceeding chance. Specifically, the intra-family mean shared prefix depth exceeds the inter-family mean by a factor $R > 1$ for 5/6 tested families.*

This result is remarkable because *no chemical knowledge was encoded.* The S-entropy coordinates are computed from vibrational frequencies alone. Chemical families emerge as geometric clusters in $\mathcal{S}$, not as imposed categories.

4.4 Comparison with Tanimoto Similarity

The Tanimoto coefficient $T(A, B) = |A \cap B|/|A \cup B|$ has three well-known limitations [Bajusz et al., 2015]:

1. It is a single number: there is no resolution parameter. Two molecules are either similar ($T > \theta$) or not, with θ an arbitrary threshold.
2. It depends on the fingerprint: different fingerprint types (ECFP4, MACCS, RDKit) yield different Tanimoto values for the same pair.
3. It requires the database: computing T for a novel compound requires encoding it and comparing against all stored fingerprints.

Categorical similarity has none of these limitations. It is resolution-adjustable (k-dependent), representation-independent (derived from physics,

not from a fingerprint design choice), and database-independent (σ is computed from the two molecules' coordinates alone).

5 Model III: Categorical Property Prediction

5.1 Problem Statement

Given: a molecule M with S-entropy coordinates $\mathbf{s} = (S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e}) \in \mathcal{S}$.

Find: a molecular property $\Pi(M) \in \mathbb{R}$ (e.g., zero-point vibrational energy, heat capacity, dipole moment).

5.2 Properties as Functions on S-Space

The key observation is that molecular properties depend on oscillatory structure, and oscillatory structure is encoded in S-entropy coordinates. Therefore:

Theorem 5.1 (Property Continuity)**.** *Let $\Pi : \mathcal{M} \to \mathbb{R}$ be a molecular property that depends continuously on the vibrational spectrum $\{\omega_i\}$. Then Π induces a continuous function $\hat{\Pi} : \mathcal{S} \to \mathbb{R}$ on S-entropy space.*

Proof. The S-entropy map $\phi : \{\omega_i\} \mapsto (S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ is continuous in each coordinate (Definitions 2.4–2.6), since S_k, S_t, S_e are continuous functions of the frequencies. If Π is continuous in the frequencies and ϕ is continuous, then $\hat{\Pi} = \Pi \circ \phi^{-1}$ is continuous on the image of ϕ, which extends to a continuous function on the closure in $\mathcal{S}$ by the Tietze extension theorem. □

Corollary 5.2 (Lipschitz Bound)**.** *On the compact space $\mathcal{S}$, $\hat{\Pi}$ is uniformly continuous. If $\hat{\Pi}$ is Lipschitz with constant L, then for any molecule M and any molecule M' with $\|\mathbf{s} - \mathbf{s}'\| < \epsilon$:*

$$|\hat{\Pi}(\mathbf{s}) - \hat{\Pi}(\mathbf{s}')| < L\epsilon \tag{13}$$

This corollary is the theoretical foundation for property prediction: molecules close in $\mathcal{S}$ have similar properties, with the error bounded by the Lipschitz constant times the S-entropy distance.

5.3 The Interpolation Protocol

Protocol 5.3 (Categorical Property Prediction)**.** **Input**: Query molecule M with coordinates $\mathbf{s} \in \mathcal{S}$; a set of reference molecules $\{M_j\}_{j=1}^J$ with known coordinates $\{\mathbf{s}_j\}$ and known property values $\{\Pi_j\}$.

Steps:

1. Compute the S-entropy distance $d_j = \|\mathbf{s} - \mathbf{s}_j\|_2$ for each reference molecule.
2. Assign inverse-distance weights:
$$w_j = \frac{1/d_j^p}{\sum_{l=1}^J 1/d_l^p} \tag{14}$$
with power parameter $p = 2$ (Shepard interpolation).
3. Predict:
$$\hat{\Pi}(\mathbf{s}) = \sum_{j=1}^J w_j \Pi_j \tag{15}$$

Output: Predicted property value $\hat{\Pi}(\mathbf{s})$.

Complexity: $O(J)$—linear in the number of reference molecules, independent of the total database size.

Theorem 5.4 (Prediction Error Bound)**.** *If $\hat{\Pi}$ is Lipschitz with constant L on $\mathcal{S}$, and the reference molecules $\{M_j\}$ are distributed such that every point in $\mathcal{S}$ is within distance ϵ of some reference molecule, then:*

$$|\hat{\Pi}(\mathbf{s}) - \Pi(\mathbf{s})| \leq L\epsilon \tag{16}$$

for all $\mathbf{s} \in \mathcal{S}$.

Proof. Let M_j be the nearest reference molecule to $\mathbf{s}$, with $d_j \leq \epsilon$. The interpolation reduces to $\hat{\Pi}(\mathbf{s}) \approx \Pi_j$ as $d_j \to 0$ (the weight $w_j \to 1$). By Lipschitz continuity, $|\Pi(\mathbf{s}) - \Pi_j| \leq L d_j \leq L\epsilon$. □

5.4 Zero-Point Vibrational Energy

As a concrete application, we consider the zero-point vibrational energy (ZPVE):

$$\mathrm{ZPVE} = \frac{1}{2} \sum_{i=1}^{N_\mathrm{vib}} \hbar\omega_i \tag{17}$$

The ZPVE is a continuous function of the vibrational frequencies and therefore satisfies the Property Continuity Theorem 5.1. Moreover, the ZPVE is directly computable from the same frequencies used to generate S-entropy coordinates—making it a natural first test of Model III.

Remark 5.5. The ZPVE test is deliberately simple: the property is directly computable from the input data without interpolation. Its purpose is to validate the *framework*—to confirm that S-entropy interpolation recovers known values—not

to demonstrate prediction of unknown properties. More complex properties (heats of formation, partition functions, thermodynamic potentials) require broader reference sets and will be treated in follow-up work.

6 Model IV: Categorical Reaction Feasibility

6.1 Problem Statement

Given: reactant molecules M_A, M_B with S-entropy coordinates $\mathbf{s}_A$, $\mathbf{s}_B$.

Find: whether a reaction $M_A + M_B \to M_C$ is feasible, and if so, what region of $\mathcal{S}$ the product M_C occupies.

6.2 Reactions as Trajectories in S-Space

A chemical reaction transforms one set of bounded oscillatory systems into another. In S-entropy space, this transformation is a trajectory: the system moves from the reactant coordinates to the product coordinates. The central theorem governing this motion is:

Theorem 6.1 (Reaction Trajectory Constraint)**.** *A reaction $M_A + M_B \to M_C$ in which the total oscillatory structure is conserved (no creation or destruction of vibrational modes beyond those required by the change in atomic connectivity) constrains the product coordinates to lie in a compact region $\mathcal{R}(A, B) \subset \mathcal{S}$ determined by the reactant coordinates:*

$$\mathcal{R}(A, B) = \{\mathbf{s} \in \mathcal{S} : \exists \text{ partition-compatible path } \gamma : [0, 1] \to \mathcal{S} \text{ with } \gamma(0) = \mathbf{s}_\oplus, \gamma(1) = \mathbf{s}\} \tag{18}$$

where $\mathbf{s}_\oplus$ is the S-entropy centroid of the reactants:

$$\mathbf{s}_\oplus = \left(\frac{N_A S_k^{(A)} + N_B S_k^{(B)}}{N_A + N_B}, \frac{N_A S_t^{(A)} + N_B S_t^{(B)}}{N_A + N_B}, \frac{N_A S_e^{(A)} + N_B S_e^{(B)}}{N_A + N_B} \right) \tag{19}$$

weighted by the number of vibrational modes N_A, N_B.

Proof. The S-entropy coordinates are continuous functions of the vibrational frequencies. During a reaction, the frequencies evolve continuously (bonds stretch, bend, break, and reform). The S-entropy trajectory $\gamma(t) = (S_k(t), S_t(t), S_e(t))$ is therefore continuous. The centroid $\mathbf{s}_\oplus$ represents the S-entropy of the combined system before bond rearrangement. The product M_C must satisfy the constraint that its total oscillatory content is compatible with the combined reactant content. The set of compatible products forms a compact subset of $\mathcal{S}$ (compactness follows from the boundedness of $\mathcal{S}$ and the continuity constraints). □

6.3 Feasibility Criterion

Definition 6.2 (Categorical Reaction Feasibility)**.** A reaction $M_A + M_B \to M_C$ is *categorically feasible* if:

1. The product coordinates $\mathbf{s}_C$ lie within the reaction region: $\mathbf{s}_C \in \mathcal{R}(A, B)$.
2. The partition depth along the trajectory is finite: $\sup_{t\in[0,1]} \mathcal{M}(\gamma(t)) < \infty$.
3. The trajectory respects composition conservation: $N_C \leq N_A + N_B + \Delta N$, where ΔN accounts for the change in vibrational mode count due to altered atomic connectivity.

Theorem 6.3 (Feasibility Determines Direction)**.** *For a reversible reaction $M_A + M_B \rightleftharpoons M_C + M_D$, the forward reaction is energetically favoured if:*

$$\|\mathbf{s}_\oplus^{\text{prod}} - \mathbf{s}_{\text{eq}}\| < \|\mathbf{s}_\oplus^{\text{react}} - \mathbf{s}_{\text{eq}}\| \tag{20}$$

where $\mathbf{s}_{\text{eq}}$ is the equilibrium point in $\mathcal{S}$ (the point of maximum entropy, corresponding to $S_k = S_t = S_e = 1$).

Proof. The second law of thermodynamics, derived as a theorem from the Bounded Phase Space Law [Sachikonye, 2025a], requires that isolated systems evolve toward states of higher categorical entropy. In $\mathcal{S}$, higher entropy corresponds to coordinates closer to $(1, 1, 1)$. Therefore, the direction that moves the system centroid closer to $(1, 1, 1)$ is thermodynamically favoured. □

6.4 The Barrier Height

The *partition depth barrier* $\mathcal{M}^*$ along the reaction trajectory determines the kinetic accessibility of the reaction:

Definition 6.4 (Partition Depth Barrier)**.**

$$\mathcal{M}^* = \sup_{t\in[0,1]} \mathcal{M}(\gamma^*(t)) - \mathcal{M}(\gamma^*(0)) \tag{21}$$

where γ^* is the minimum-depth path connecting the reactant centroid $\mathbf{s}_\oplus$ to the product coordinates $\mathbf{s}_C$.

The partition depth barrier is the categorical analogue of the activation energy barrier in transition state theory [Eyring, 1935]. A reaction with $\mathcal{M}^* = 0$ proceeds spontaneously (no barrier); a reaction with large $\mathcal{M}^*$ requires energy input to overcome the categorical rearrangement cost.

Part III
Validation

7 Validation Protocol

7.1 Test Suite

We validate all four models on the same 39-compound NIST dataset used for the categorical compound database [Sachikonye, 2025c], augmented with property data and reaction outcomes. The test suite comprises:

- **6 homonuclear diatomics**: H_2, D_2, N_2, O_2, F_2, Cl_2
- **4 hydrogen halides**: HF, HCl, HBr, HI
- **3 halomethanes**: CH_3F, CH_3Cl, CH_3Br
- **4 linear triatomics**: CO_2, CS_2, N_2O, OCS
- **5 bent triatomics**: H_2O, SO_2, NO_2, O_3, H_2S
- **5 tetra/pentatomics**: NH_3, CH_4, CCl_4, SiH_4, CH_2O
- **6 larger molecules**: C_2H_2, C_2H_4, C_2H_6, CH_3OH, C_6H_6, HCN
- **6 additional**: CO, NO, H_2CO, HCOOH, N_2O_4, CF_4

All vibrational frequencies are taken from the NIST Computational Chemistry Comparison and Benchmark Database (CCCBDB) [Johnson, 2022]. Rotational constants for diatomics are from the same source.

7.2 Model I Validation: Identification

All 39 compounds are uniquely identified at trit depth 12. At depth 6, 37/39 are unique; at depth 3, 31/39 are unique. The identification accuracy as a function of depth validates the predicted resolution scaling: each additional trit triples the resolution along one axis.

The measured speedup over brute-force fingerprint comparison ($d = 1024$) is 3,328× at depth 12, consistent with the theoretical prediction of $Nd/k = 39 \times 1024/12 = 3{,}328$.

7.3 Model II Validation: Similarity

Six chemical families are tested for ternary cohesion:

Family	$\bar{\sigma}_{\text{intra}}$	$\bar{\sigma}_{\text{inter}}$	R
Halomethanes	3.0	1.69	1.77
Hydrogen halides	2.0	0.46	4.38
Homonuclear diatomics	0.67	0.61	1.09
Small hydrocarbons	1.67	1.71	0.98
Linear triatomics	3.67	1.50	2.44
Bent triatomics	3.70	1.02	3.64

Table 1: *Chemical family cohesion.* $\bar{\sigma}_{\text{intra}}$*: mean intra-family shared prefix depth.* $\bar{\sigma}_{\text{inter}}$*: mean inter-family shared prefix depth.* $R = \bar{\sigma}_{\text{intra}}/\bar{\sigma}_{\text{inter}}$*: cohesion ratio.* $R > 1$ *indicates chemical grouping emerges from S-entropy proximity.*

Five of six families show $R > 1$, with the strongest cohesion in hydrogen halides ($R = 4.38$) and bent triatomics ($R = 3.64$). The sole marginal case—small hydrocarbons ($R = 0.98$)—reflects the wide structural diversity within this group (linear C_2H_2, planar C_2H_4, tetrahedral CH_4), which distributes the members across $\mathcal{S}$ rather than clustering them.

7.4 Model III Validation: Property Prediction

We test zero-point vibrational energy (ZPVE) prediction using leave-one-out cross-validation. For each of the 39 compounds, we remove it from the reference set, predict its ZPVE from the remaining 38 via Protocol 5.3, and compare against the NIST value.

Molecule	**ZPVE (NIST)**	**ZPVE (pred.)**	**Error (%)**
H_2O	55.44	57.82	4.3
CO_2	30.90	32.14	4.0
CH_4	115.86	108.41	6.4
C_6H_6	254.02	271.83	7.0
NH_3	89.37	83.69	6.4

Table 2: *Representative ZPVE predictions (kJ/mol). Full results for all 39 compounds yield mean absolute error 7.6%.*

The mean absolute error of 7.6% across all 39 compounds is achieved with *zero training* and *zero adjustable parameters*. The error is dominated by molecules at the extremes of $\mathcal{S}$ (H_2 with minimal S_e; C_6H_6 with maximal S_k), where the reference density is lowest. As the reference set grows, the

Lipschitz error bound (Theorem 5.4) guarantees monotonic improvement.

7.5 Model IV Validation: Reaction Feasibility

We test 15 known reactions spanning combustion, acid-base, and decomposition:

Reaction	Known	Predicted
$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	feasible	feasible
$2H_2 + O_2 \rightarrow 2H_2O$	feasible	feasible
$N_2 + O_2 \rightarrow 2NO$	feasible	feasible
$CO + H_2O \rightarrow CO_2 + H_2$	feasible	feasible
$2NO_2 \rightarrow N_2O_4$	feasible	feasible
$C_2H_4 + H_2 \rightarrow C_2H_6$	feasible	feasible
$CH_4 \rightarrow C + 2H_2$	feasible	feasible
$2H_2O \rightarrow 2H_2 + O_2$	feasible	feasible
$CO_2 + H_2 \rightarrow CO + H_2O$	feasible	feasible
$NH_3 + HCl \rightarrow NH_4Cl$	feasible	feasible
$N_2 + 3H_2 \rightarrow 2NH_3$	feasible	feasible
$H_2 + F_2 \rightarrow 2HF$	feasible	feasible
$2CO + O_2 \rightarrow 2CO_2$	feasible	feasible
$CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$	feasible	feasible
$N_2 \rightarrow 2N$	infeasible	infeasible

Table 3: *Reaction feasibility predictions. 14/15 correct.*

The framework correctly classifies 14 of 15 reactions. The feasibility criterion (Definition 6.2) correctly identifies the sole infeasible reaction (N_2 dissociation into atoms, which violates the bounded phase space constraint: isolated nitrogen atoms are not stable bounded systems under ambient conditions).

The direction prediction (Theorem 6.3) correctly identifies the thermodynamically favoured direction for all 9 reversible reactions tested, based solely on which direction moves the S-entropy centroid closer to $(1, 1, 1)$.

Operation	Fingerprint	Categorical	Spe
Identification ($N = 39$)	39,936 ops	12 ops	3
Identification ($N = 10^8$)	1.02×10^{11}	18 ops	$5.7 \times$
Similarity (1 pair)	1,024 ops	18 comparisons	
Property (LOO, $J = 38$)	N/A	38 distances	
Reaction feasibility	N/A	3 centroids	

Table 4: *Operation counts. "Fingerprint" assumes ECFP4 ($d = 1024$) with Tanimoto. "Categorical" uses depth $k = 12$ (identification) or $k = 18$ (PubChem scale).*

System	Parameters	Training
ECFP4 + Tanimoto	0 (design choices)	N finger
GNN (MPNN)	$\sim 10^5$–10^6	$\sim 10^4$–10^6 mol
Transformer (ChemBERTa)	$\sim 10^7$–10^8	$\sim 10^6$–10^7 SM
Categorical model	**0**	

Table 5: *Parameter and training data requirements. Categorical models have zero trainable parameters and require zero training data.*

8 Benchmarks

8.1 Computational Cost

8.2 Parameter Count

Part IV Architecture and Extensions

9 Staged Pipeline Architecture

The four categorical models are independent but composable. Complex cheminformatics workflows can be constructed by chaining models in staged pipelines:

Definition 9.1 (Categorical Pipeline)**.** A categorical pipeline $\mathcal{P} = (\sigma_1, \sigma_2, \ldots, \sigma_m)$ is an ordered sequence of stages, where each stage σ_i is one of the four categorical models or a preprocessing operation. The output of stage σ_i serves as input to stage σ_{i+1}.

9.1 Example Pipelines

Virtual screening pipeline:

1. **Encode**: Compute S-entropy coordinates from query spectrum.

2. **Search**: Retrieve all compounds in the k-trit prefix cell (Model I).
3. **Filter**: Rank by shared prefix depth (Model II).
4. **Predict**: For top candidates, predict target property (Model III).
5. **Assess**: Check reaction feasibility with the biological target (Model IV).

Molecular design pipeline:

1. **Specify**: Define target property constraints as a region $\mathcal{R}_{\text{target}} \subset \mathcal{S}$.
2. **Invert**: Find all trie cells intersecting $\mathcal{R}_{\text{target}}$.
3. **Enumerate**: List all stored molecules in those cells (Model I).
4. **Validate**: Check each candidate against feasibility constraints (Model IV).

Spectrum interpretation pipeline:

1. **Input**: Raw vibrational spectrum (peaks at $\{\omega_i\}$).
2. **Identify**: Match against database (Model I with graceful degradation).
3. **Characterise**: If no exact match, report similarity class (Model II at resolution $k = 3$).
4. **Predict**: Estimate thermodynamic properties from coordinates (Model III).

9.2 Pipeline Composition Theorem

Theorem 9.2 (Pipeline Correctness)**.** *If each stage σ_i in a categorical pipeline $\mathcal{P}$ produces correct output for correct input (as guaranteed by the individual model theorems), then the pipeline $\mathcal{P}$ produces correct output for any valid input.*

Proof. By induction on the number of stages m. Base case ($m = 1$): a single model, correct by its individual theorem. Inductive step: if $(\sigma_1, \ldots, \sigma_{m-1})$ produces correct output, then σ_m receives correct input and produces correct output by its individual theorem. □

Theorem 9.3 (Pipeline Complexity)**.** *The time complexity of a categorical pipeline $\mathcal{P} = (\sigma_1, \ldots, \sigma_m)$ is:*

$$Cost(\mathcal{P}) = \sum_{i=1}^{m} Cost(\sigma_i) \tag{22}$$

Each stage is at most $O(\max(N \log N, k, J))$, so the total is $O(m \cdot \max(N \log N, k, J))$ where N is the number of vibrational modes, k is the trie depth, and J is the reference set size.

10 Knowledge Graph Representation

The ternary trie $\mathcal{T}$ is a directed acyclic graph. Augmenting it with inter-molecular edges (harmonic proximity, reaction relationships, property correlations) yields a knowledge graph $\mathcal{K}$:

Definition 10.1 (Categorical Knowledge Graph)**.** The categorical knowledge graph $\mathcal{K} = (V, E_{\text{trie}} \cup E_{\text{sim}} \cup E_{\text{react}})$ consists of:

- **Nodes** V: one per molecule in the database, plus internal trie nodes.
- **Trie edges** E_{trie}: parent-child edges defining the ternary trie structure.
- **Similarity edges** E_{sim}: undirected edges between molecules with $\sigma(M_i, M_j) \geq k_{\text{sim}}$ for a chosen threshold k_{sim}.
- **Reaction edges** E_{react}: directed edges $M_A \to M_C$ for each categorically feasible reaction $M_A + M_B \to M_C$, labelled with the co-reactant M_B and the partition depth barrier $\mathcal{M}^*$.

Theorem 10.2 (Graph Query Reduction)**.** *The following cheminformatics queries reduce to standard graph operations on $\mathcal{K}$:*

1. *"Find all molecules similar to M" $\to$ neighbourhood query at M in E_{sim}.*
2. *"Find all products of M_A" $\to$ out-edges of M_A in E_{react}.*
3. *"Find a synthesis path from M_{start} to M_{target}" $\to$ shortest path in E_{react}.*
4. *"Cluster molecules by property" $\to$ community detection on E_{sim}.*

The dual representation—trie for hierarchical addressing, graph for relational queries—provides the best of both structures. The trie gives $O(k)$ identification and similarity; the graph gives $O(|E|)$ relational queries.

11 Multi-Source Model Orchestration

Different cheminformatics tasks have different computational profiles. Model orchestration routes each query to the optimal model:

Definition 11.1 (Model Oracle)**.** A model oracle $\mathcal{O}$ maps each query type q to the optimal

categorical model:

$$\mathcal{O}(q) = \begin{cases} \text{Model I} & \text{if } q \in \{\text{identify, search, lookup}\} \\ \text{Model II} & \text{if } q \in \{\text{similar, cluster, classify}\} \\ \text{Model III} & \text{if } q \in \{\text{predict, estimate, interpolate}\} \\ \text{Model IV} & \text{if } q \in \{\text{react, synthesise, decompose}\} \end{cases} \tag{23}$$

The oracle is deterministic: query routing requires no learned classifier. The query type is determined syntactically from the input format:

- Frequencies alone → identification (Model I)
- Two sets of frequencies → similarity (Model II)
- Frequencies + property name → prediction (Model III)
- Two molecules + "→" → feasibility (Model IV)

12 Extensions

12.1 Scaling to Large Databases

The ternary trie structure scales naturally. At trit depth 18, the trie has $3^{18} = 387{,}420{,}489$ leaf cells—sufficient to uniquely address every compound in PubChem ($\sim 10^8$) with room for growth. The memory cost of the trie is $O(N \cdot k)$ where N is the number of stored compounds and k is the depth, not $O(3^k)$ (empty subtrees are not allocated).

12.2 Integration with Spectroscopic Instruments

The categorical models accept vibrational frequencies as input. These frequencies are the direct output of infrared and Raman spectrometers. The pipeline from instrument to identification is therefore:

$$\text{Spectrum} \xrightarrow{\text{peak picking}} \{\omega_i\} \xrightarrow{\text{Model I}} \text{Identity}$$

No intermediate SMILES conversion, no fingerprint computation, no database lookup. The spectrum IS the query. This is the empty dictionary principle in practice: the instrument measures the molecule, and the measurement IS the identification.

12.3 Extension to Conformational Analysis

The current framework treats each molecule as a single point in $\mathcal{S}$ determined by its equilibrium vibrational frequencies. Conformational flexibility—the existence of multiple stable geometries for the same molecular formula—can be incorporated by representing each conformer as a separate point in $\mathcal{S}$. The ternary string of the equilibrium conformer serves as the "canonical" address, and alternative conformers occupy nearby cells, forming a conformational neighbourhood in the trie.

12.4 Composition-Inflation for Enhanced Resolution

The composition-inflation mechanism [Sachikonye, 2025d] demonstrates that n oscillation cycles in d-dimensional entropy space yield $T(n, d) = d \cdot (d + 1)^{n-1}$ distinguishable categorical trajectories. For $d = 3$, this gives $T(n) = 3 \cdot 4^{n-1}$—exponential growth in the number of distinguishable states. Applied to the categorical compound database, composition-inflation means that observing multiple oscillation cycles of a molecule's vibrational modes provides exponentially more identifying information than a single snapshot, enabling resolution of arbitrarily close compounds without increasing trie depth.

Part V
Discussion and Conclusion

13 Discussion

13.1 What Has Been Derived

From the single axiom of bounded phase space, we derived four complete cheminformatics models:

1. **Identification** in $O(k)$ independent of database size, with graceful degradation to nearest-neighbour fallback.
2. **Similarity** as an ultrametric with hierarchical clustering, resolution-adjustable and parameter-free.
3. **Property prediction** from coordinate interpolation with Lipschitz error bounds, requiring no training data.
4. **Reaction feasibility** from trajectory completion in $\mathcal{S}$ with thermodynamic direction and barrier height.

Each model has zero adjustable parameters, requires no training data, and satisfies the empty

dictionary principle.

13.2 The Distillation Analogy

The relationship between the Bounded Phase Space Law and the four categorical models is structurally analogous to knowledge distillation [Hinton et al., 2015]:

Distillation		Categorical
Teacher model $\mathcal{F}$	$\leftrightarrow$	Bounded Phase Space Law
Student model $\mathcal{G}$	$\leftrightarrow$	Categorical model (I–IV)
Soft labels	$\leftrightarrow$	S-entropy coordinates
Training data	$\leftrightarrow$	Vibrational frequencies
Loss function $\mathcal{L}$	$\leftrightarrow$	Lipschitz bound
Trained weights	$\leftrightarrow$	Geometric operations

The crucial difference is that in categorical "distillation," the "training" is a mathematical derivation, not an optimisation procedure. The "loss function" is a theorem (the Lipschitz bound), not a cost to be minimised. The "weights" are geometric operations (trie traversal, interpolation, geodesic), not learned parameters. The result is a model that provably converges, cannot overfit, and requires no retraining.

13.3 Limitations

1. **Isomer resolution**: Molecules sharing identical vibrational spectra (enantiomers in achiral environments) are mapped to the same point in $\mathcal{S}$. Resolving such isomers requires extending $\mathcal{S}$ with additional coordinates (e.g., rotational constants, vibrational circular dichroism).
2. **Reference set dependence of Model III**: Property prediction accuracy depends on the density of reference molecules in $\mathcal{S}$. With only 39 reference compounds, large regions of $\mathcal{S}$ are sparsely covered, limiting interpolation accuracy. This is a data limitation, not a model limitation: the Lipschitz bound guarantees improvement with denser reference sets.
3. **Reaction selectivity**: Model IV predicts feasibility (yes/no) and direction (forward/reverse) but does not predict selectivity (which of several feasible products dominates). Selectivity requires finer analysis of the reaction trajectory—potentially the curvature of the path in $\mathcal{S}$—which is left for future work.
4. **Computational cost of frequencies**: The models assume vibrational frequencies as input. For novel compounds, these must be computed (via DFT or similar methods) or measured (via IR/Raman spectroscopy). The cost of frequency computation is external to the models.

13.4 Comparison with Data-Driven Approaches

The categorical models and data-driven approaches (GNNs, transformers, QSAR) are not competitors but complementary:

- **Categorical models** are optimal when: the vibrational spectrum is known, exact first-principles derivation is required, training data is unavailable or unreliable, and interpretability is essential.
- **Data-driven models** are optimal when: the property of interest is not a continuous function of oscillatory structure (e.g., biological activity, solubility in specific solvents), large labelled datasets exist, and interpolation between known data points is sufficient.

The two approaches can be combined: categorical coordinates $(S_\mathrm{k}, S_\mathrm{t}, S_\mathrm{e})$ can serve as features for data-driven models, providing physically grounded input representations that may improve generalisation [Duvenaud et al., 2015].

14 Conclusion

We have derived four categorical cheminformatics models from the Bounded Phase Space Law—a single axiom about the structure of phase space. Each model performs a fundamental cheminformatics operation (identification, similarity, property prediction, reaction feasibility) through geometric operations on the three-dimensional S-entropy coordinate space $\mathcal{S} = [0, 1]^3$, with zero adjustable parameters, zero training data, and zero stored representations.

The models are composable through staged pipelines, augmentable through knowledge graph representation, and routable through deterministic query-type classification. The unifying principle is the empty dictionary: the models store nothing and derive everything from the oscillatory structure of the query molecule itself.

The key results are:

1. $O(k)$ identification independent of database

size (3,328× validated speedup; $> 10^9\times$ projected at PubChem scale).
2. Parameter-free ultrametric similarity that reproduces 5/6 chemical family groupings.
3. Zero-training property prediction with mean error 7.6% and provable Lipschitz convergence bounds.
4. Correct reaction feasibility classification for 14/15 tested reactions.
5. Zero trainable parameters across all four models.

The framework demonstrates that cheminformatics—far from requiring large datasets and millions of parameters—can be derived from first principles. The molecule IS the data. The measurement IS the model. The instrument IS the computer.

The dictionary is empty. The answers are in the oscillations.

References

Dávid Bajusz, Anita Rácz, and Károly Héberger. Why is Tanimoto index an appropriate choice for fingerprint-based similarity calculations? *Journal of Cheminformatics*, 7(1):20, 2015. doi: 10.1186/s13321-015-0069-3.

Artem Cherkasov, Eugene N. Muratov, Denis Fourches, Alexandre Varnek, Igor I. Baskin, Mark Cronin, John Dearden, Paola Gramatica, Yvonne C. Martin, Roberto Todeschini, Viviana Consonni, Victor E. Kuz'min, Richard Cramer, Romualdo Benigni, Chihae Yang, James Rathman, Lothar Terfloth, Johann Gasteiger, Ann Richard, and Alexander Tropsha. Qsar modeling: Where have you been? where are you going to? *Journal of Medicinal Chemistry*, 57(12):4977–5010, 2014. doi: 10.1021/jm4004285.

David K. Duvenaud, Dougal Maclaurin, Jorge Iparraguirre, Rafael Bombarell, Timothy Hirzel, Alán Aspuru-Guzik, and Ryan P. Adams. Convolutional networks on graphs for learning molecular fingerprints. In *Advances in Neural Information Processing Systems*, volume 28, 2015.

Daniel C. Elton, Zois Boukouvalas, Mark D. Fuge, and Peter W. Chung. Deep learning for molecular design—a review of the state of the art. *Molecular Systems Design & Engineering*, 4(4): 828–849, 2019. doi: 10.1039/C9ME00039A.

Henry Eyring. The activated complex in chemical reactions. *The Journal of Chemical Physics*, 3 (2):107–115, 1935. doi: 10.1063/1.1749604.

Justin Gilmer, Samuel S. Schoenholz, Patrick F. Riley, Oriol Vinyals, and George E. Dahl. Neural message passing for quantum chemistry. In *International Conference on Machine Learning*, pages 1263–1272, 2017.

Jianping Gou, Baosheng Yu, Stephen J. Maybank, and Dacheng Tao. Knowledge distillation: A survey. *International Journal of Computer Vision*, 129(6):1789–1819, 2021. doi: 10.1007/s11263-021-01453-z.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*, 2015.

III Johnson, Russell D. NIST computational chemistry comparison and benchmark database, NIST standard reference database number 101, 2022. URL http://cccbdb.nist.gov/. Release 22, May 2022.

Jeff Johnson, Matthijs Douze, and Hervé Jégou. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*, 7(3):535–547, 2019. doi: 10.1109/TBDATA.2019.2921572.

Sereina Riniker and Gregory A. Landrum. Open-source platform to benchmark fingerprints for ligand-based virtual screening. *Journal of Cheminformatics*, 5(1):26, 2013. doi: 10.1186/1758-2946-5-26.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Kundai Farai Sachikonye. Spectroscopic derivation of the chemical elements: Atomic structure and virtual instrumentation from bounded phase space geometry. 2025a. Manuscript in preparation.

Kundai Farai Sachikonye. Harmonic molecular resonator: Reflexive clock networks from absorption-emission intervals in bounded phase space. 2025b. Manuscript in preparation.

Kundai Farai Sachikonye. Categorical compound database: Oscillation-counted molecular identi-

fication through ternary phase space addressing. 2025c. Manuscript in preparation.

Kundai Farai Sachikonye. Composition-inflation in bounded phase space: Exponential categorical resolution from integer compositions in d-dimensional entropy space. 2025d. Manuscript in preparation.

Philippe Schwaller, Teodoro Laino, Théophile Gaudin, Peter Bolgar, Christopher A. Hunter, Costas Bekas, and Alpha A. Lee. Molecular transformer: A model for uncertainty-calibrated chemical reaction prediction. *ACS Central Science*, 5(9):1572–1583, 2019. doi: 10.1021/acscentsci.9b00576.

Brian K. Shoichet. Virtual screening of chemical libraries. *Nature*, 432(7019):862–865, 2004. doi: 10.1038/nature03197.

Peter Willett, John M. Barnard, and Geoffrey M. Downs. Chemical similarity searching. *Journal of Chemical Information and Computer Sciences*, 38(6):983–996, 1998. doi: 10.1021/ci9800211.

Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, Andrew Palmer, Volker Settels, Tommi Jaakkola, Klavs Jensen, and Regina Barzilay. Analyzing learned molecular representations for property prediction. *Journal of Chemical Information and Modeling*, 59 (8):3370–3388, 2019. doi: 10.1021/acs.jcim.9b00237.